

HAT-P-20b

R-1398

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_151210 · created 2026-08-01T03:48:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457366.80391 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.011
Transit depth (Rp/Rs)^2	2.83 +/- 0.37 [%]
Orbital Inclination (inc)	86.65 +/- 0.46
Airmass coefficient 1 (a1)	1.0288 +/- 0.0097
Airmass coefficient 2 (a2)	-0.024 +/- 0.0064
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#1 - [233, 219]
Optimal Aperture	3.46
Optimal Annulus	16.45
Transit Duration (day)	0.0794 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.011 vs 0.1284 (deviation 3.6 σ)
Duration fit vs published	0.0794 d vs 0.0775 d (deviation 0.35 σ)
Mid-time O-C	-1.5 min
Depth SNR	15.3
Residual scatter	0.95 %

Light curve

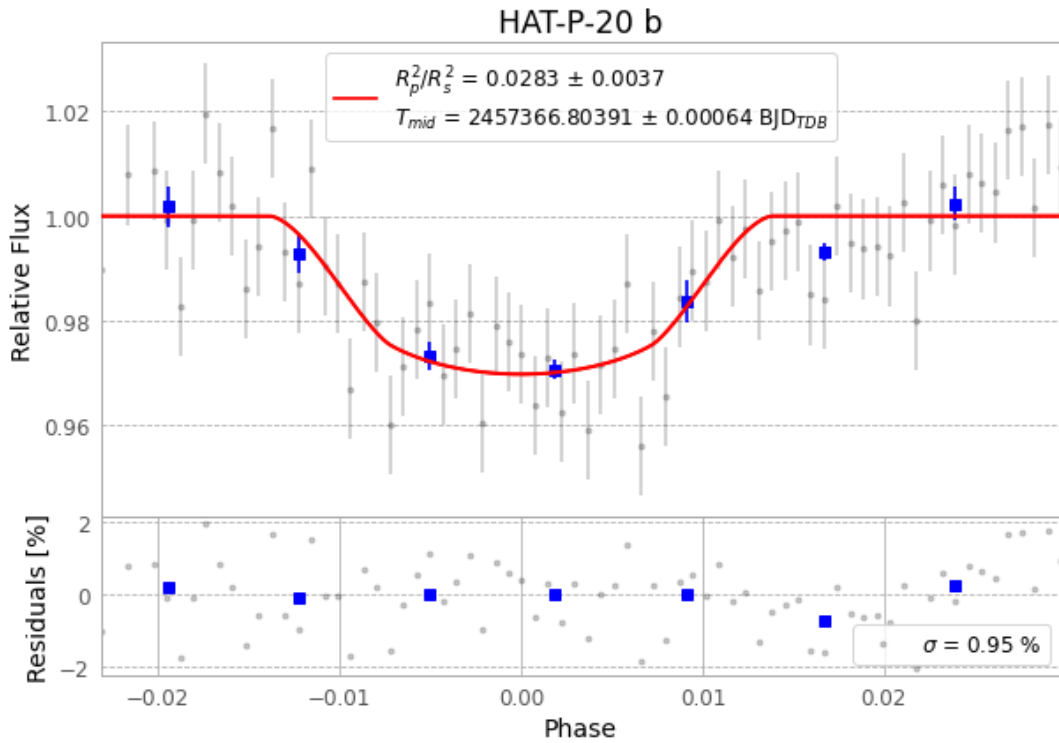


Figure 1 — FinalLightCurve_HAT-P-20 b_2015-12-09.png (SHA-256 6d9b5d09f28be20e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [120, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 761d2f0c0da3b3cd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), HATP-20151210052412.FITS → HATP-20151210091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-151210.log (f44272c55676...), FinalLightCurve_HAT-P-20 b_2015-12-09.png (6d9b5d09f28b...), FinalLightCurve_HAT-P-20 b_2015-12-09.csv (3f780cc9fddf...)

Compute resources

Wall time 380.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)